

ARUN SANYASI

Instrumentation and Control Engineer | PLC/SCADA | Electrical Automation | Hydropower Automation
Samdrup Jongkhar, Bhutan | +975 77694520 | sanyasiarun220@gmail.com | Portfolio: Arun.github

CAREER OBJECTIVE

Seeking a Graduate Engineer / PLC-SCADA / Electrical Automation role to apply hydropower automation, commissioning, instrumentation, control-panel and electrical maintenance skills in reliable industrial plant operations.

EDUCATION

B.E. Instrumentation and Control Engineering
College of Science and Technology, RUB **2022–2026**
Result: **68.21% till 7th semester**

Class XII, Karmaling Higher Secondary School **2020–2021**
Result: **70.2%**

Class X, BCSE **2019**
Result: **78.14%**

EXPERIENCE

Bhutan Automation and Engineering Limited **12 Weeks**
Graduate Engineer Trainee – Automation, SCADA and Commissioning

- Completed hydropower OJT covering SCADA integration, commissioning and HT/LT panel assembly.
- Developed inlet, outlet and flushing-gate logic with manual, automatic, remote and local modes.
- Worked with Siemens SICAM A8000 AI, DI, DO, AO modules and SCALA 250 signal mapping.
- Interpreted EPLAN schematics for signal tracing, ferruling, labelling and wiring verification.
- Assisted ACB/MCCB installation, internal wiring, busbar installation and final inspection support.

Druk Wang Alloys Limited **Dec 2024–Jan 2025**
Instrumentation and Electrical Automation Intern

- Gained exposure to switchyard, MCCs, motor-control wiring, protection systems and 33kV/11kV systems.
- Observed capacitor banks, preventive maintenance practices and sensor-to-panel signal tracing.

Fuzzy Automation **Jul 2025**
Automation and Digitization App Intern

- Supported handwritten bills and receipts digitization workflow, digital archiving and capture-to-storage testing.

TECHNICAL SKILLS

PLC/SCADA: Siemens SICAM A8000, Siemens S7-1200, SCALA 250, TIA Portal, WinCC, Factory I/O, CODESYS, FluidSIM

Control/Commissioning: Ladder Logic, Structured Text, PID, interlocks, alarms, loop checks, I/O verification, signal mapping

Electrical Systems: HT/LT panels, MCCs, ACB/MCCB, busbars, motor-control wiring, capacitor banks, VFD wiring, 33kV/11kV exposure

Design/Programming: EPLAN Electric P8, AutoCAD Electrical, Proteus, Arduino IDE, Python, C, C++, SolidWorks, Fusion 360

Protocols/Protection: Modbus RTU/TCP, PROFINET, PROFIBUS, SIPROTEC, ABB Relion, GE Multilin

ROLE FIT

- Suitable for Graduate Engineer, PLC/SCADA Trainee, Instrumentation Trainee and Electrical Automation roles.
- Ready to support hydropower automation, substations, ferro-alloy plants, utilities and process industries.
- Able to assist in commissioning, loop checks, signal tracing, control-panel verification and site documentation.

LANGUAGES

English | Dzongkha | Hindi | Lhotshamkha

PROJECTS

Basic Health Monitoring Station: Final-year project measuring BMI, pulse, SpO₂, respiratory rate, temperature, height and weight with GUI and QR report.

PLC Dam Level Control: PLC-based dam water-level monitoring and control logic with alarm and control-decision functions.

SICAM A8000 Boiler Monitoring: SICAM Workbench and SCALA 250 monitoring for boiler temperature, pressure and level loops.

Factory I/O + CODESYS Line: Simulated conveyor sorting, filling, capping, storage and HMI-based bottle-production workflow.

Smart Accident Detector: Designed accident detection and alerting concept for Bhutanese road conditions.

Power-System Case Studies: Completed solar microgrid, AI-based EV battery management and solid-state transformer studies.

ACHIEVEMENTS AND LEADERSHIP

- IEEE publication: *Design and Development of a Basic Health Monitoring Station*.
- Participated in Student Research Meet organised by Royal University of Bhutan.
- Class Representative for B.E. Year 1 and Year 2.
- Served as Public Address Councilor for college communication and events.
- Winner of college-level Business Idea Competition while representing CST.
- Founder of Zumthruel, a virtual reality gaming venture.
- Active contributor to Entrepreneur Club, GovTech Satellite Program, Vocational Club and Integrity Club.

WORKSHOPS AND TRAINING

- PLC Simulator Factory.
- VFD Control Wiring Diagram Techniques – 179 lessons.
- Electrical Testing Equipment Techniques.
- Oil and Gas Industry Fundamentals.
- Robotic Process Automation for Developers.
- Selenium with Python from Scratch.
- AI Industry Analysis and Business Entrepreneurship.
- Hydropower automation and commissioning exposure at Dagachhu Hydropower Plant.

SOFT SKILLS

- Field troubleshooting and practical problem solving.
- Technical documentation and reporting discipline.
- Team coordination and communication.
- Fast learning in industrial plant environments.
- Safety awareness during field and electrical work.
- Leadership through class representation and project coordination.

REFERENCES

Deepan Sharma – CEO, Bhutan Automation and Engineering Ltd.
+975 17641448 | ceo@bhutanautomation.com

Ashika Chettri – Control Engineer, Bhutan Automation and Engineering Ltd.
+975 77936493

Karma Kelzang Eudon – Faculty, College of Science and Technology
+975 17607817

Mani Kumar Powdyel – Faculty Member, College of Science and Technology
+975 17369937

DECLARATION

I hereby declare that the information provided above is true and complete to the best of my knowledge.

Place: Samdrup Jongkhar, Bhutan

Signature: Arun Sanyasi